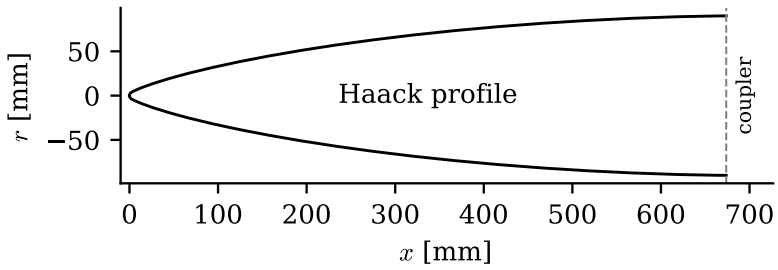
Weisshorn nose cone at 720 mm: geometry sheet

2027_C_FD_NOSE-CONE-720MM_GEOMETRY Fabio Feig (FD), 29.10.2026

Design A: LV-Haack 677 + 43 mm (baseline, 3U payload inside the nose)

x is measured along the axis from the blunted tip, r is the outer radius, all lengths in mm.



$$\begin{aligned}
&\text{Spherical tip} && 0 \le x \le 2.420: && r = \sqrt{r_n^2 - (r_n - x)^2} \\
&\text{Haack profile} && 2.420 \le x \le 673.700: && r = \frac{R}{\sqrt{\pi}}\sqrt{\theta - \frac{\sin 2\theta}{2} + C\sin^3\theta}, \qquad \theta = \arccos\!\left(1 - \frac{2\,(x+\delta)}{L}\right) \\
&\text{Coupler cylinder} && 673.700 \le x \le 716.700: && r = R
\end{aligned}$$

Parameter	Value
Haack constant C	1/3
Curved length to the sharp apex L	677.000
Body radius R	90.000 ($D = 180$)
Tip radius r_n	4.500
Length removed by blunting δ	3.300
Tangency point (x, r)	(2.420, 3.991)
Curved part, tip to shoulder	673.700
Coupler cylinder c	43.000
Total physical length	**716.700**

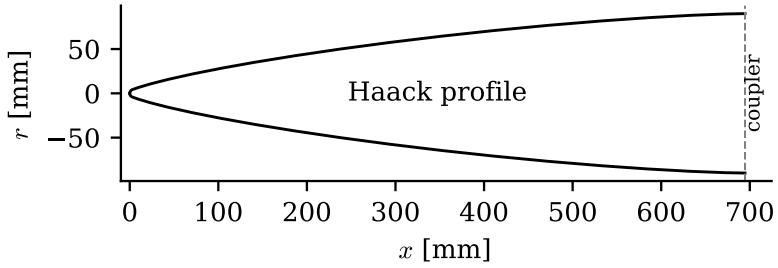
The profile is tangent to the sphere at the tangency point and to the cylinder at the shoulder ($\mathrm{d}r/\mathrm{d}x = 0$).

CAD: `nose_lvhaack_L677_c43_rn4.5mm.step` (solid), `.stl`, `.csv` (profile points).

x	r	x	r
0.000	0.000	375.000	74.139
2.420	3.991	400.000	76.444
25.000	13.083	425.000	78.563
50.000	20.756	450.000	80.500
75.000	27.324	475.000	82.260
100.000	33.176	500.000	83.847
125.000	38.490	525.000	85.262
150.000	43.366	550.000	86.508
175.000	47.870	575.000	87.585
200.000	52.046	600.000	88.489
225.000	55.927	625.000	89.214
250.000	59.540	650.000	89.742
275.000	62.903	673.700	90.000
300.000	66.032	675.000	90.000
325.000	68.941	700.000	90.000
350.000	71.640	716.700	90.000

Design B: Von Kármán 700 + 20 mm (backup, payload elsewhere)

x is measured along the axis from the blunted tip, r is the outer radius, all lengths in mm.



$$\begin{aligned}
&\text{Spherical tip} && 0 \le x \le 2.905: && r = \sqrt{r_n^2 - (r_n - x)^2} \\
&\text{Haack profile} && 2.905 \le x \le 694.600: && r = \frac{R}{\sqrt{\pi}}\sqrt{\theta - \frac{\sin 2\theta}{2} + C\sin^3\theta}, \qquad \theta = \arccos\!\left(1 - \frac{2\,(x+\delta)}{L}\right) \\
&\text{Coupler cylinder} && 694.600 \le x \le 714.600: && r = R
\end{aligned}$$

Parameter	Value
Haack constant C	0
Curved length to the sharp apex L	700.000
Body radius R	90.000 ($D = 180$)
Tip radius r_n	4.500
Length removed by blunting δ	5.400
Tangency point (x, r)	(2.905, 4.208)
Curved part, tip to shoulder	694.600
Coupler cylinder c	20.000
Total physical length	**714.600**

The profile is tangent to the sphere at the tangency point and to the cylinder at the shoulder ($\mathrm{d}r/\mathrm{d}x = 0$).

CAD: `nose_vonkarman_L700_c20_rn4.5mm.step` (solid), `.stl`, `.csv` (profile points).

x	r	x	r
0.000	0.000	375.000	67.062
2.905	4.208	400.000	69.734
25.000	11.082	425.000	72.278
50.000	17.285	450.000	74.693
75.000	22.725	475.000	76.979
100.000	27.678	500.000	79.132
125.000	32.274	525.000	81.148
150.000	36.584	550.000	83.019
175.000	40.655	575.000	84.736
200.000	44.518	600.000	86.286
225.000	48.196	625.000	87.647
250.000	51.705	650.000	88.787
275.000	55.058	675.000	89.644
300.000	58.264	694.600	90.000
325.000	61.330	700.000	90.000
350.000	64.262	714.600	90.000